Atreay Kukanur

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.34
Delhi Public School, Ruby Park, Kolkata	Class XII (CBSE)	2019	90.00%
Bodhicarya Senior Secondary School, New Town, Kolkata	Class X (CBSE)	2017	9.60%

WORK EXPERIENCE

Full Stack Developer | Builderz | Remote *(Jan 2024 - Jun 2024)*

- Developed a real-time, role-based chat platform using **Next.js**, **React**, **PubNub**, and **Firebase** supporting live data synchronization.
- Integrated a **Solana** backend into the chat platform, implementing decentralized access, authentication, and cloud data persistence.

MASTER'S THESIS

SCOPE: Structure-aware Components Optimization for Phishing Detection in Ethereum Networks *(Ongoing)*

- Enhanced Ethereum phishing detection by designing graph representation learning techniques using **ANA, BiGIS, temporal multigraph modeling**, and **self-supervised pretraining**.
- Developed an end-to-end graph learning pipeline from raw blockchain transactions to phishing classification with improved **temporal feature engineering** and **graph pretraining**.
- Improved benchmark performance to **69.12% F1** on **BERT4ETH** (exceeding the original paper's **65-67% F1**) and achieved **93.94% F1** on **MulDiGraph**.

PROJECTS

MLOpsForge - Model-Agnostic ML Infrastructure Platform - GitHub

- Built a production-grade **MLOps platform** supporting end-to-end model training, experiment tracking, model registry, deployment and live inference on a **SLURM GPU cluster**.
- Engineered a **plugin-based architecture** supporting Graph Neural Networks, CNNs and State-Space Models without modifying platform code.
- Integrated **MLflow, Kafka, Prometheus, Grafana** and a React dashboard for experiment monitoring, real-time job tracking and low-latency model serving.

FlowForge - Distributed AI Workflow Orchestration Engine - GitHub

- Designed a distributed **DAG-based workflow orchestration engine** featuring a workflow compiler, task scheduling and dependency validation for ML/LLM pipelines.
- Implemented **multiple scheduling algorithms**, distributed task queues, checkpoint recovery and heartbeat-based fault tolerance for reliable workflow execution.
- Built an observability stack using **Prometheus, Grafana** and Docker/SLURM deployment supporting scalable execution across distributed workers.

ResearchAgent - Multi-Agent AI Research Assistant - GitHub

- Built a **multi-agent research system** implementing planning, literature retrieval, synthesis and evaluation using the Model Context Protocol (MCP).
- Integrated semantic search, persistent memory and self-reflection with a self-hosted **ServeLLM backend** for autonomous research workflows.
- Optimized concurrent document retrieval and ranking, reducing literature search latency while improving retrieval relevance.

ZK-Whistle - Anonymous Whistleblowing Platform - GitHub

- Built a decentralized **anonymous whistleblowing platform** using **Semaphore Zero-Knowledge Proofs**, enabling verified group members to submit reports without revealing their identities.
- Implemented **Ethereum smart contracts**, IPFS-based evidence storage and a gasless **relayer architecture** that prevents wallet-address linkage to whistleblowing transactions.
- Developed a full-stack application using **Next.js, Solidity, Hardhat and Ethers.js**, providing secure report submission and a real-time on-chain feed of ZK-verified reports.

POSITIONS OF RESPONSIBILITY

Outreach Coordinator: IIT Hyderabad Alumni Cell - Coordinating student-alumni engagement initiatives through institute outreach programs like **IITH Alumni Legacy Meet 2026** where we managed **200+ attendees** (Jul 2026 - Present).

Captain and Volunteer: National Service Scheme (NSS) - Led and Contributed with around **250+ student volunteers** in organizing community service initiatives like **Cleanliness Drive**, **Old Age Home Visit**, **Government School Visit**, etc (Sep 2019 - Apr 2022).

SKILLS

Programming Languages: Python, C++, C, JavaScript, SQL, Bash. **AI / Machine Learning:** Machine Learning, Deep Learning, Generative AI, LLM, Agentic AI, RAG, NLP, Computer Vision. **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Transformers, LangChain, LangGraph, FastAPI. **Tools & Platforms:** Git, Linux, Docker, Interactive Notebook, SLURM GPU Clusters, REST APIs. **Systems:** Linux, Bash, Docker, SLURM Workload Manager, Linux Command Line, Virtual Environments.

RELEVANT COURSES

Deep Learning | Foundations of Machine Learning | Advanced Data Structures and Algorithms | Operating Systems | Database Management Systems | Linear Optimization | Program Analysis and Optimization | Visual Computing | Fraud Analytics using Predictive and Social Network Techniques

SCHOLASTIC ACHIEVEMENTS

- **LFX Mentorship(CNCF Org)** - Past Mentee and Open-source contributor
- **ETHIndia 2023** - Selected among top ~6%-8% of applicants and a Finalist
- **TimeChain Summer of Code 2023** - Selected as contributor to india's 1st blockchain based open-source contributor program.
- **BuildSpace N&W S4 Builder 2023** - Selected for a 6-week, YC & a16z-funded program to develop solutions for complex challenges.
- **Competitive Programming** - Currently Codeforces Expert (1812); 430 LeetCode problems; 400+ Code360 Problems

HOBBIES AND INTEREST

- **Sports & Fitness:** Training my own physical model at the gym, or debugging via basketball and badminton.
- **Interests:** Processing an expanding dataset of 50+ anime series and manga to prevent mental overfitting.